



Topologically-Constrained Multi-Task Learning for Periodontitis Grading

Cholponai Manapova

Computer Vision

Sapienza University of Rome

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Problem Statement



Periodontitis is a common dental disease that causes progressive bone loss around teeth. Manual grading from radiographs is time-consuming and depends on clinical expertise.

The **goal** of this project is to automatically analyze dental radiographs and predict disease severity.

Goal

Develop a deep learning model for:



Segmentation



Periodontitis Grading

State of the Art

CNN classification

- Disease grading from radiographs

U-Net segmentation

- Medical image segmentation

Mask R-CNN

- Instance segmentation

Multi-task learning

- Joint segmentation and classification

Challenges:

- Class imbalance
- Limited annotations
- Anatomical inconsistency

DenPAR Dataset



650

Training samples

150

Validation samples

200

Testing samples

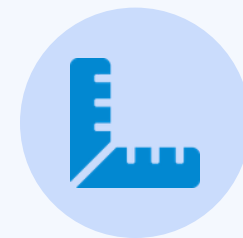
Available Annotations



Radiograph-wise masks



Tooth-wise masks

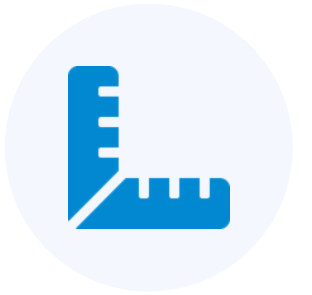


CEJ keypoints



Bone-level annotations

Data Processing



CEJ–Bone Distance

distance < 20

$20 \leq \text{distance} < 50$

distance ≥ 50

Grade 0

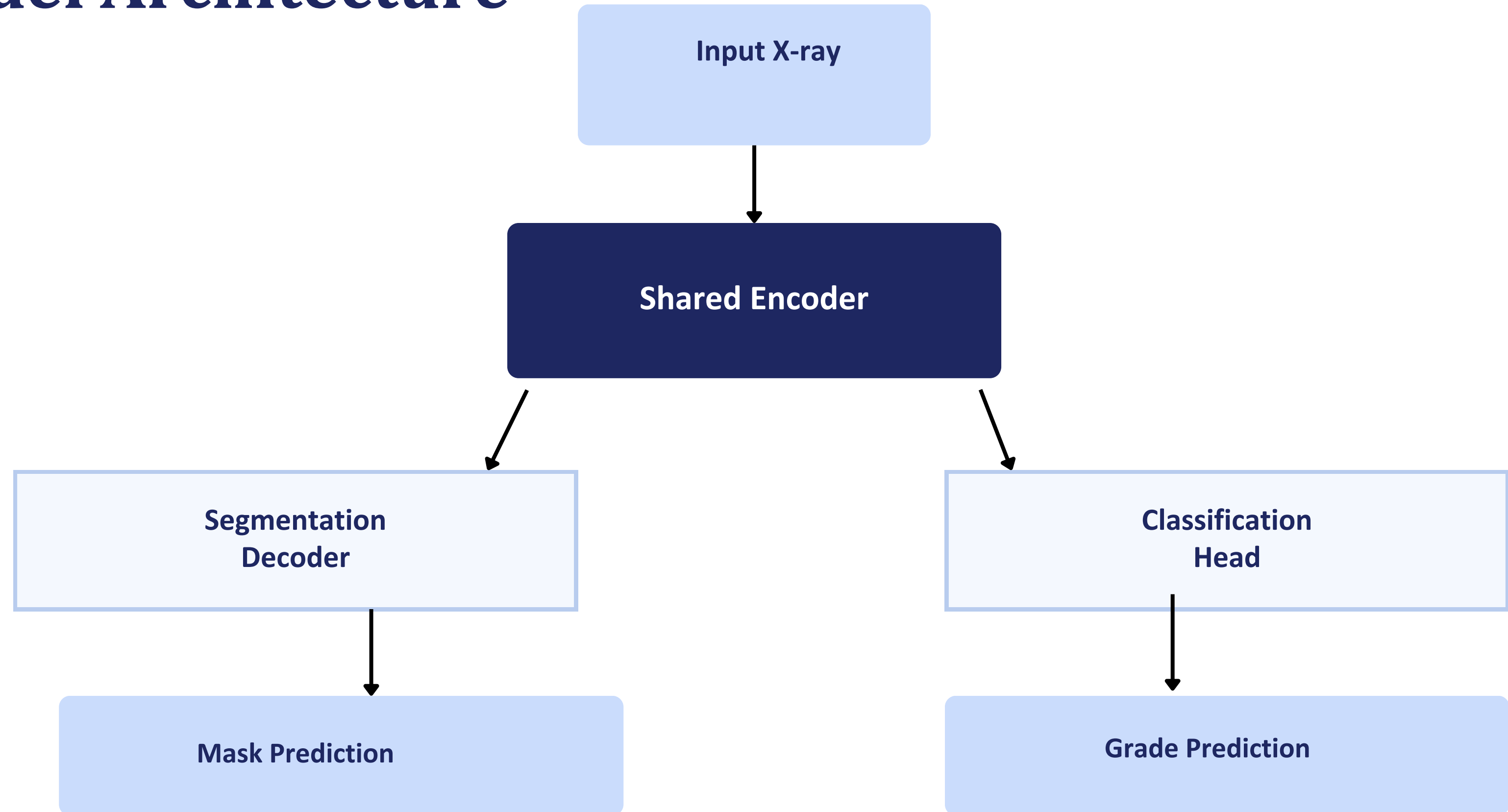
Grade 1

Grade 2

Weak labels were automatically generated from anatomical measurements.

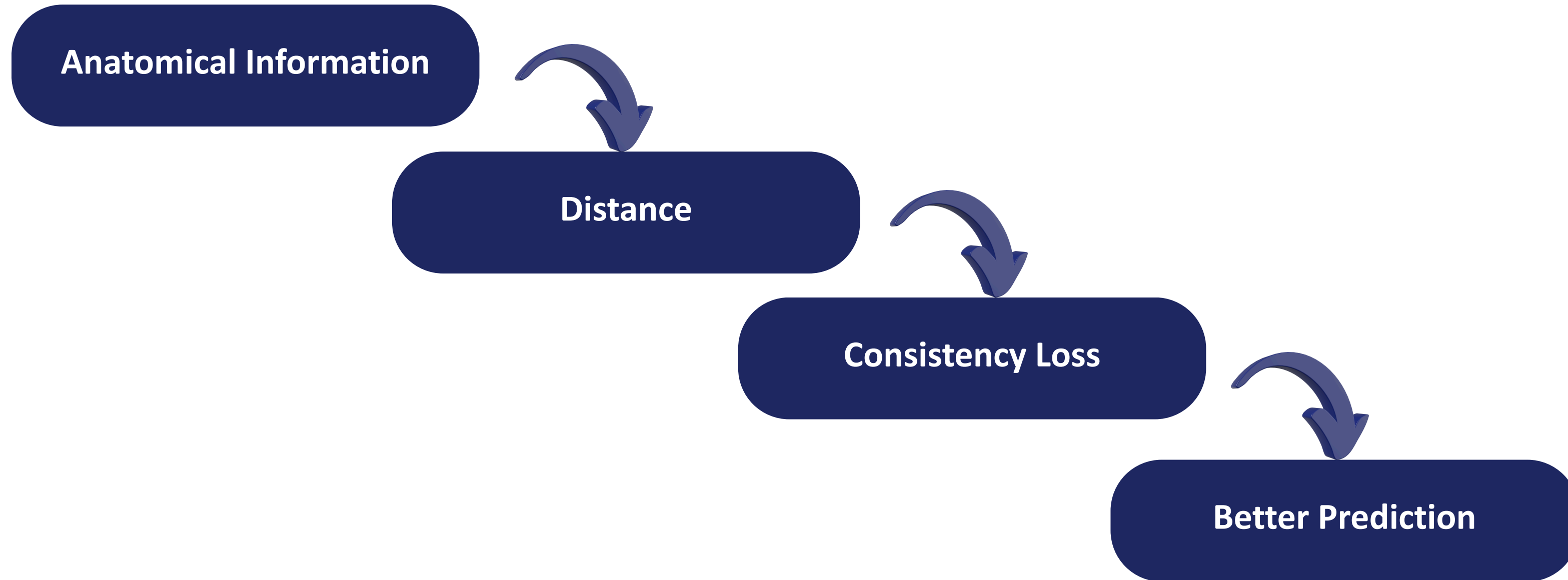
Model Architecture

Multi-Task U-Net



Proposed Method

Anatomical Consistency Loss



The proposed loss introduces anatomical knowledge into training and encourages clinically meaningful predictions

Experimental Setup

PyTorch

T4 GPU
Google Colab

4

2
Batch
Size

0.0001

LR

256x256

Training & Image
Size

4

Epochs

8

Results



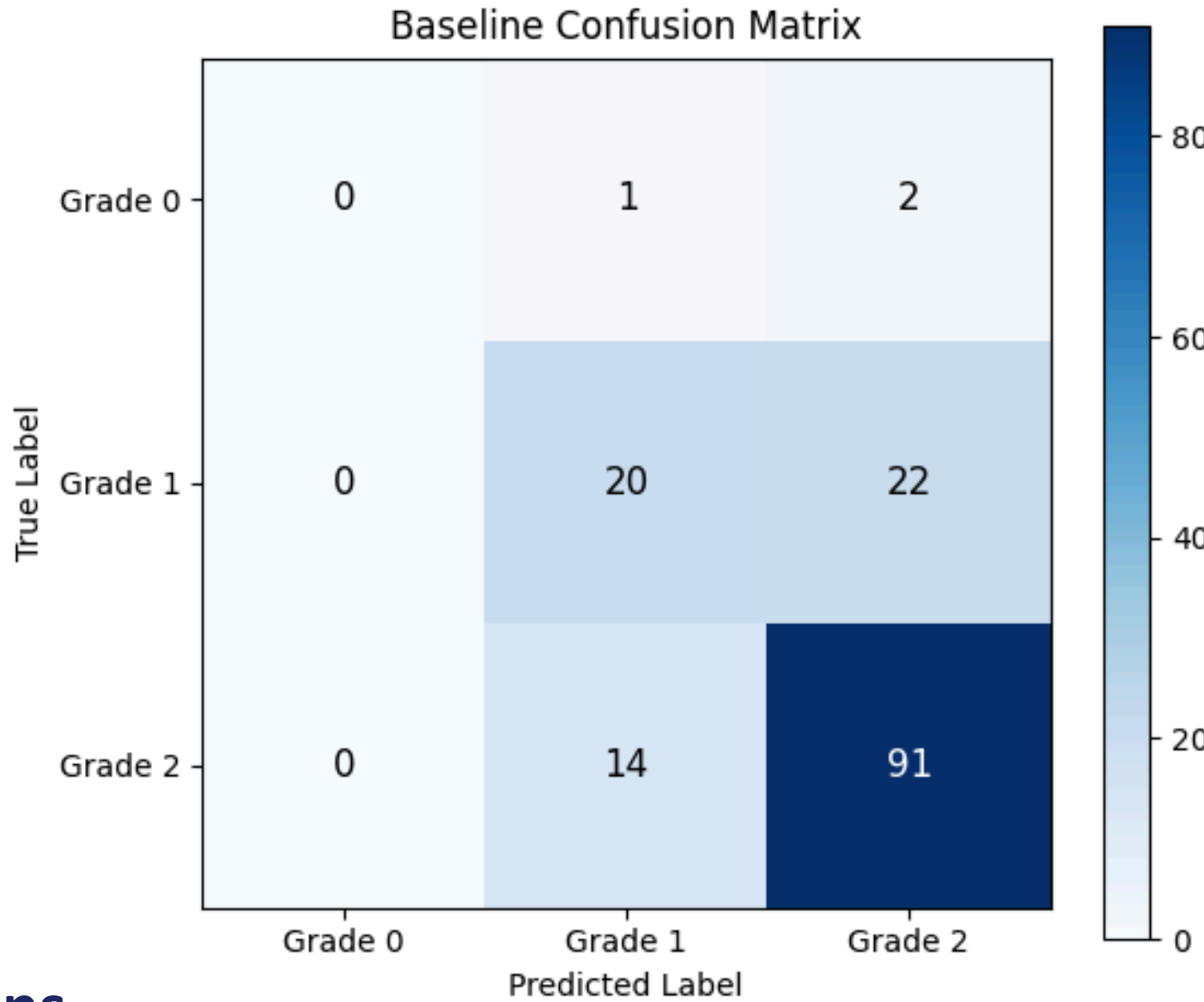
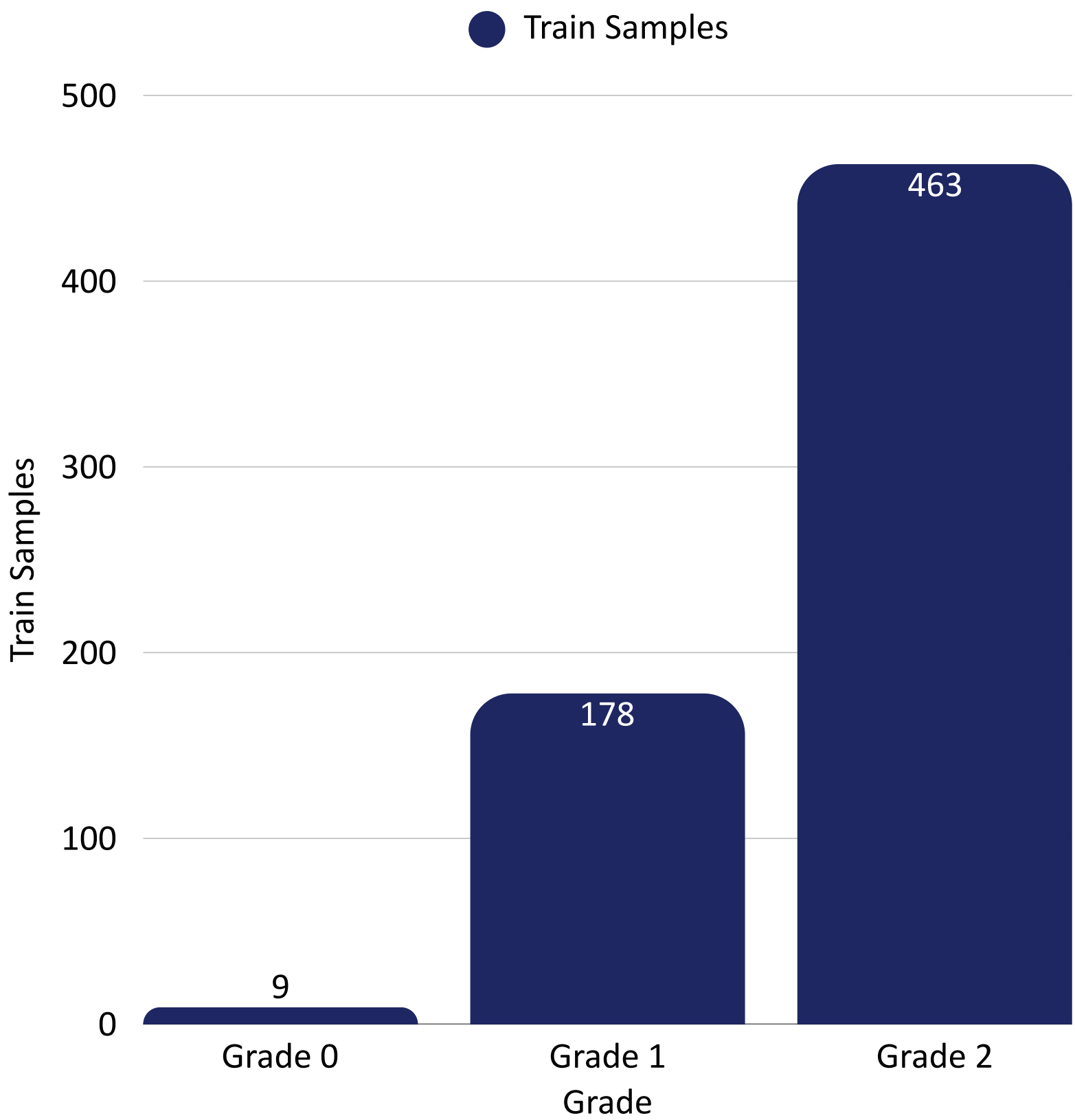
Metric	Baseline	Proposed
Accuracy	74.0%	70.0%
Dice	0.906	0.895
IoU	0.840	0.823



Important

Dataset is highly imbalanced.

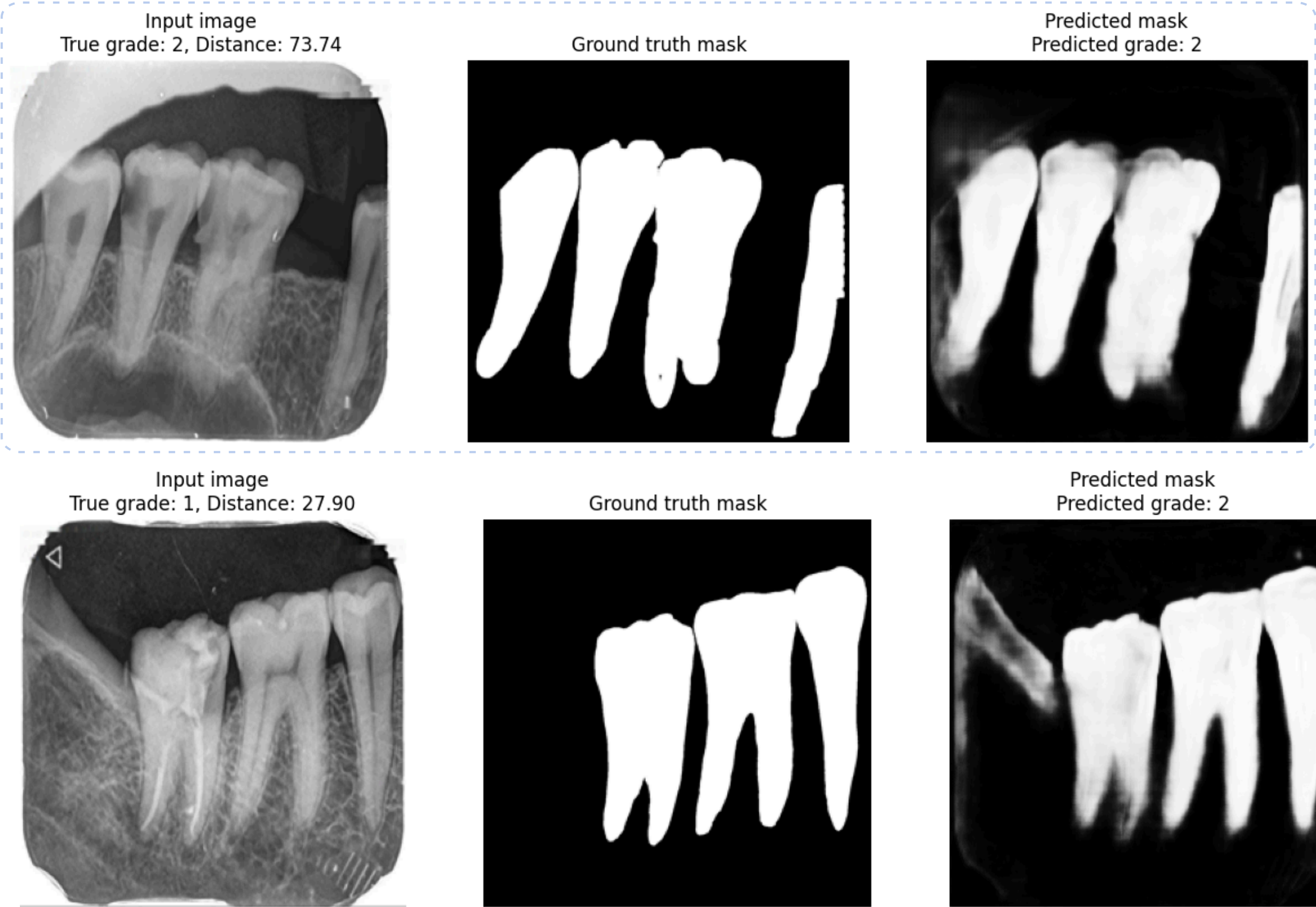
Why the model makes mistakes (Error Analysis)



Key Observations

- Grade 2 is recognized most accurately.
- Grade 1 is sometimes confused with Grade 2.
- Grade 0 is difficult to classify.
- Severe class imbalance affects performance.

Qualitative Results

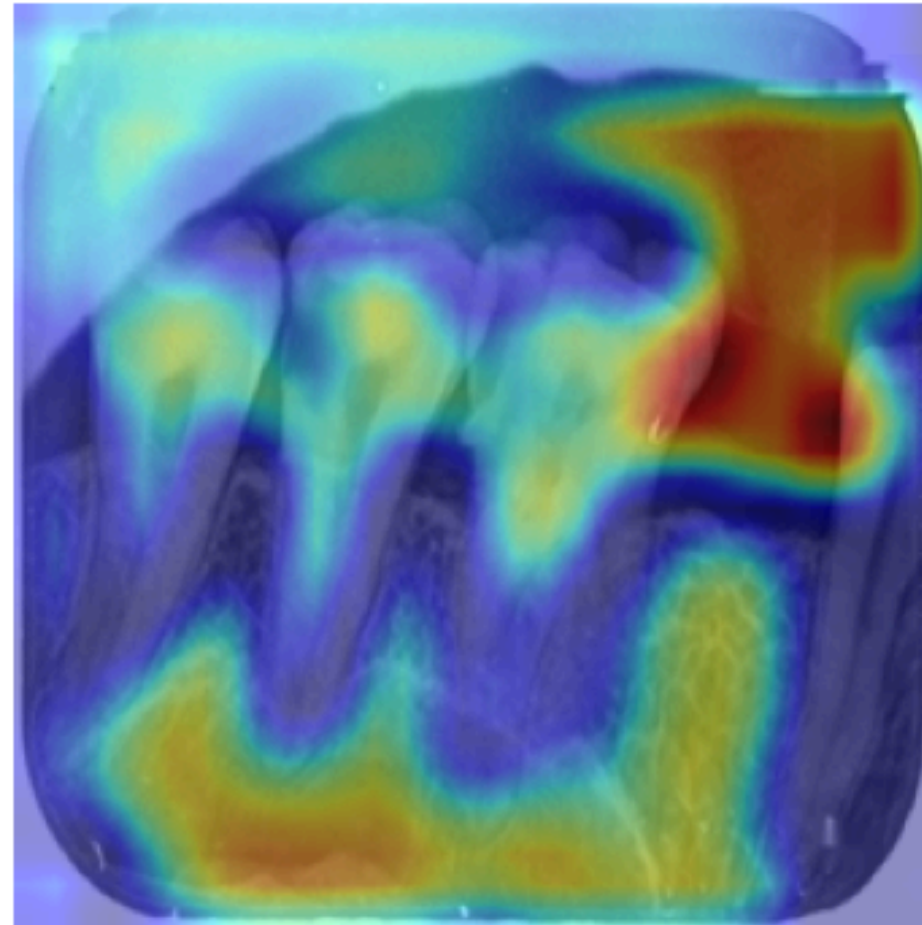


Grad-CAM

Input Image
True Grade: 2



Grad-CAM
Predicted Grade: 2



Ground Truth Mask



Grad-CAM highlights the image regions that influenced the classification decision.

This helps verify whether the model focuses on clinically relevant anatomical areas.



Conclusion

- Multi-task learning was successfully implemented
- Anatomical Consistency Loss was introduced
- Baseline achieved the best performance: Accuracy = 74.0%
- Grad-CAM was used to analyze model interpretability

Future Work

Expert clinical labels

Better class balancing

More training epochs



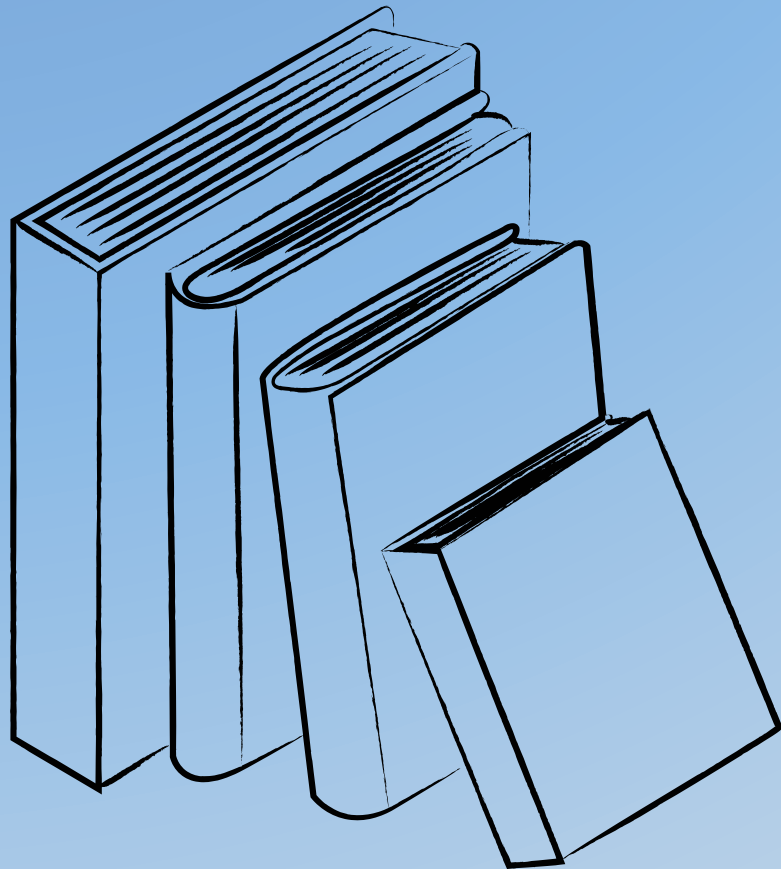
References

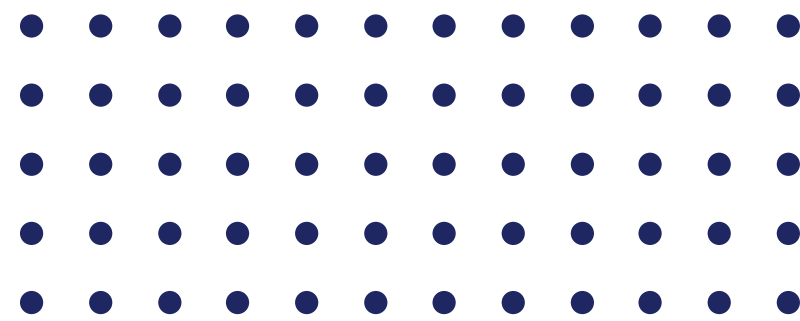
DenPAR Dataset

Ronneberger et al. (2015), U-Net

PyTorch Documentation

Computer Vision Course Material





THANK YOU

Presentation by Cholponai Manapova

